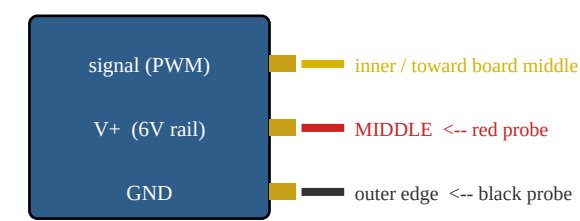


Servo-Power Measurement Worksheet

6DOF arm · PCA9685 servo rail investigation · full log: hardware/bench-log-servo-power.md

Rule: 12 V OFF for continuity tests (B, C) — ON for voltage tests (D). All 16 middle pins share one V+ rail; any channel is representative. A healthy copper path beeps **continuously**; a beep that fades after ~1 s is the capacitor charging, not continuity.



one channel column, viewed from the servo side

	Measurement	Meter / power	Probe points	Expected	RESULT
A	Photo: channel banks + terminal block, straight down (pin-base colors readable)	camera	—	row order confirmed	
B	Continuity: GND path	beep · 12V OFF	GND terminal screw ↔ edge pin (any channel)	solid continuous beep	
C	Continuity: V+ path — THE VERDICT	beep · 12V OFF	V+ terminal screw ↔ middle pin (any channel)	solid beep = healthy; fading beep = V+ trace BREAK (fault found)	
D	Servo rail voltage	DC V · 12V ON	black on GND screw · red pressed on middle pin	+6.0 V	
E	Photo: probes in place during C or D	camera	—	placement verified	
F	Servo test (only after D reads 6 V)	—	servo on channel 0, brown lead at board edge; then swap servo	motion on 'w' wiggle	
G	If C fades: inspect terminal-block solder tabs (board underside), reflow, re-run C	iron	—	solid beep after reflow	
H	Common-ground integrity (logic vs servo rail)	DC V · 12V ON, then beep · OFF	ESP32 GND pin ↔ PCA9685 GND terminal screw	~0 V (<50 mV) and solid beep; jumpy reading = bad ground path (also explains ESP32 resets)	

Notes / readings: